

Samuel Alvarez

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EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Science, Minor in Data and Society

Fall 2022 - Present

- Relevant Coursework: Algorithms, Databases, Software Engineering, Principles of Data and Society, Data and Society in Practice
- Dean's List Fall 2022 – Spring 2025

PROJECTS

Post Office Database

Jan 2025 – May 2025

HTML/CSS, JavaScript, SQL, Node.js, VS Code

- Developed a full-stack web application in a 5-person team to simulate and manage post office functions, including package tracking, customer data management, and sales reporting.
- Engineered automated sales reports with SQL queries and dynamic dashboards, enabling managers to track weekly revenue and identify service trends.
- Collaborated in designing the entity-relationship (ER) diagram for the database, ensuring efficient data structure and normalization.
- Designed the website interface with a clean and intuitive layout to enhance user experience.

Obesity in Houston Research Proposal

Aug 2024 – Dec 2024

- Reviewed Houston's ongoing public health efforts to reduce obesity and identified gaps in current approaches.
- Created a directed acyclic graph (DAG) to illustrate relationships between socioeconomic factors such as median income and obesity prevalence.
- Collected and processed datasets from the US Census, CDC, and Harris County Appraisal District to support statistical analysis.
- Drafted a structured data analysis plan outlining regression tests and correlation studies to evaluate trends.
- Presented findings and next-step recommendations to faculty advisors, highlighting data limitations and areas for further study.

Pharis Fellowship Education Reform Research Proposal

Aug 2023 – Dec 2023

- Collaborated with a 4-member team to research Texas education policy and propose reforms targeting STAAR standardized testing.
- Analyzed TAPR datasets to evaluate Houston ISD performance from 2018–2022, identifying patterns and shortcomings in test outcomes.
- Created data visualizations and geocoded maps to highlight inequities in test improvement measurement.
- Presented findings on systemic issues such as unequal district funding, low graduation rates, flawed standardized testing, and declining student mental health.

EXPERIENCE

Math Instructor

Aug 2023 – Present

Huntington Learning Center, Clear Lake

Houston, TX

- Delivered individualized instruction for K-12 math and standardized test prep (SAT, ACT, ASVAB), improving student comprehension and performance.
- Adapted teaching methods to meet diverse learning styles, balancing conceptual explanations with practical applications to build confidence and retention.
- Helped students progress from failing to excelling academically, with measurable outcomes including SAT score increases of up to 100 points.

TECHNICAL SKILLS

Languages: Python, C++, HTML/CSS, JavaScript, SQL, R

Frameworks: Node.js

Developer Tools: Git, VS Code, RStudio